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| **Fiscal Year:** | FY 2013 | **Task Last Updated:** | FY 07/19/2012 |
| **PI Name:** | Beckermann, Christoph Ph.D. | | |
| **Project Title:** | Effect of Convection on Columnar-to-Equiaxed Transition in Alloy Solidification | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | MATERIALS SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Materials science | | |
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| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
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| **PI Email:** | becker@engineering.uiowa.edu | **Fax:** | FY 319-335-5669 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 319-335-5681 |
| **Organization Name:** | University of Iowa | | |
| **PI Address 1:** | Department of Mechanical Engineering | | |
| **PI Address 2:** | 2402 Seamans Center | | |
| **PI Web Page:** | http://www.engineering.uiowa.edu/ | | |
| **City:** | Iowa City | **State:** | IA |
| **Zip Code:** | 52242-1527 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2010 Materials Science NNH10ZTT001N |
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| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA MSFC |
| **Contact Monitor:** | Downey, James1 Patton | **Contact Phone:** | |
| **Contact Email:** | patton.downey@nasa.gov | | |
| **Flight Program:** | | | |
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| **Performance Goal Text:** | | | |
| | The objective of the proposed research is to develop computational models of the columnar-to-equiaxed transition (CET) in alloy solidification, in order to help define and interpret previous and future microgravity experiments performed under the ESA-sponsored CETSOL project. The CETSOL team recently completed a batch of six microgravity experiments using the MSL-LGF facility onboard the ISS. Additional microgravity experiments are planned that employ the MSL-SQF and MSG-DIRSOL facilities.<br>Gaining a more complete understanding of the complex physical phenomena accompanying the CET is important for predicting the grain structure of castings. Open scientific questions include the role played by melt convection, fragmentation of dendrite arms, and the transport of fragments and equiaxed crystals in the melt. Terrestrial experiments performed by the CETSOL team show that these gravity-induced phenomena have a strong effect on the CET, and the | | |

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| **Task Description:** | relative motion of solid and liquid results in strong macrosegregation. The microgravity experiments, on the other hand, provide benchmark data for the case where melt convection and solid transport are absent.<br><br>The proposed research will utilize models at three different length scales: phase-field, mesoscopic, and volume-averaged models. The phase-field model will be needed to resolve the growth and transport processes at the scale of the microstructure, the mesoscopic model will allow for simulations at the scale of individual grains, while the volume-averaged model will be used to perform simulations of the entire experiments. In order to simulate the associated terrestrial experiments, all three of the proposed models will include melt convection and transport of solid. A new model will be developed for the fragmentation process. Detailed comparisons will be performed between the model predictions and the experimental data. Based on the outcome of the research, new microgravity CET experiments will be proposed. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The columnar-to-equiaxed transition (CET) in the grain structure of metal alloy castings has fascinated researchers in the solidification area for more than 50 years. The CET refers to the transition between the elongated grains in the outer portions of a casting and the more rounded grains in the center. Understanding this transition is fundamental to determining what type of grain structure forms in castings of most metal alloys (steel, aluminum, copper, etc.). Often, a fully equiaxed structure is preferred, but the fully columnar structures of many turbine blades are an important exception. In addition to its high practical significance, the CET represents a “holy grail” in the area of modeling and simulation of casting. This is because in order to realistically predict the CET, almost every physical phenomenon at every length scale must be taken into account simultaneously: heat transfer, solute transport, melt flow and the transport of small dendrite fragments and equiaxed grains on the scale of the casting; the thermal/solutal/mechanical interactions between the growing grains/dendrites; and the nucleation of grains (especially in the presence of grain refiners) and fragmentation of existing dendrites. The research will not only provide an improved understanding of the CET, but also models and computer simulations of the grain structure formation in metal castings that can be used by industry to better understand and optimize their casting processes. |
| **Task Progress:** | Progress was made in all areas of the research on the columnar-to-equiaxed transition (CET) in alloy solidification: phase-field simulation, mesoscopic simulation, macroscopic simulation, collaboration with the European CETSOL team, and completing a Science Requirement Document for future microgravity experiments on the CET.<br>Three-dimensional phase-field simulations of alloy solidification are being conducted to study the dendrite fragmentation process on a microscopic (microstructure) scale. Fragmented dendrite sidebranches are believed to be a potent source of equiaxed grains. For this purpose, simulations are being conducted for columnar dendritic growth with an imposed temperature gradient and cooling rate. After a fully dendritic structure is obtained, the cooling rate is suddenly reduced. This leads to fragmentation of the dendrites at the junction between primary and secondary sidebranches. The fragmentation dynamics and rates are being studied as a function of the growth conditions. Considerable effort was devoted to parallelizing the code in order to allow for large scale simulations to be conducted. Speedups by more than a factor of 50 have been achieved. In addition, the deformation of dendrites is being studied using phase-field simulations. Such deformation can also lead to fragmentation. During the next project year, these simulations will be continued. A theory of fragmentation will be developed.<br><br>Mesoscopic simulations of columnar and equiaxed solidification are being performed in order to investigate in detail the evolution of the grain structure on an intermediate scale. In this type of simulation, the evolution of the dendrite envelopes is tracked, while the solute field is calculated only in the extra-dendritic space between the envelopes. A three-dimensional computer code has been written and preliminary simulations have been performed to compare the predicted envelope shapes with available measurements. The next step is to include melt convection. This work is well underway.<br><br>Macroscopic simulations are being conducted to study the CET on the scale of an entire casting. A previously developed volume-averaged model is used for these simulations. The governing equations are solved using the public domain OpenFoam CFD software platform. The code was tested for columnar and equiaxed solidification without melt convection and transport of solid. Work is underway to include gravity-driven convection and grain sedimentation/floatation. Ultimately, macroscopic simulations will be conducted to help analyze and design past and future microgravity experiments.<br><br>The PI participated in an ESA CETSOL project meeting in December 2011 at Brunel University in London, UK. During this meeting, the progress of the European team in analyzing results from recently completed ISS experiments was reviewed. In addition, detailed discussions were held to plan future microgravity experiments regarding the CET. These interactions have now resulted in a concrete experimental plan. The next CETSOL meeting will be help in Aachen, Germany in September 2012.<br><br>A draft of a Science Requirement Document was prepared for future microgravity CET experiments on the ISS. The hypothesis of the experiments is that fragmentation of columnar dendrites can lead to a CET in non-grain-refined alloys. As opposed to current theories, the experiments will show that a growth velocity increase and nucleation of equiaxed grains ahead of the columnar front are not necessary conditions for a CET to occur. It is proposed to conduct solidification experiments where a CET is induced by a growth velocity decrease, rather than an increase. The Low Gradient Furnace (LGF) on the MSL will be used for the proposed experiments. Experiments are proposed for various Al-Cu and Al-Si alloys. All CET experiments will consist of two solidification phases (which are preceded by a homogenization phase). During the first phase, a columnar grain structure will be established by imposing a certain temperature gradient and cooling rate (furnace translation rate). During the second phase, the portion of the sample that is still in mushy and liquid states should be located in the hot zone of the furnace and be held adiabatically, while all heat extraction occurs from the cold zone through the already solidified portion. No further furnace translation is desired. The imposition of adiabatic conditions along the sample that is still in a fully or partially liquid state will prevent additional nucleation of solid. During the second phase, the columnar front velocity will first experience a growth velocity increase until all remaining melt superheat is dissipated. Then, the columnar front will slow down because it grows into an isothermal undercooled liquid. This growth velocity decrease will induce fragmentation of the columnar grains. Because the entire melt is undercooled, the fragments will not re-melt and a CET can occur. Other details of the proposed experiments can be found in the draft of the Science Requirement Document. |

| Bibliography Type: | Description: (Last Updated: 12/04/2024) |
|---|---|
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